Kalman Filter Forecast
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The examples below use the `*KFAS*` package (*Helske*, 2017), the `*dlm*` package (*Petris et al*., 2009), and the `*forecast*` package. All packages are available on CRAN.

Example 1: Local Level Model (Nile River Data)

The local level model (also called the random walk plus noise model) is the simplest non-trivial SSM. It decomposes a series into a stochastic level μ_t and an irregular component:

$$y_t = \mu_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2)$$

$$\mu_{t+1} = \mu_t + \eta_t, \qquad \eta_t \sim \mathcal{N}(0, \sigma_\eta^2)$$

Here $\mathbf{Z}_t = 1, \mathbf{T}_t = 1, \mathbf{R}_t = 1, \mathbf{H}_t = \sigma_\varepsilon^2, \mathbf{Q}_t = \sigma_\eta^2$.

The signal-to-noise ratio $q = \sigma_\eta^2/\sigma_\varepsilon^2$ controls the smoothness of the estimated level: when $q = 0$, the level is constant; when $q \to \infty$, the filtered level tracks observations exactly.

Example 2: Local Linear Trend Model

The local linear trend model extends the local level by adding a stochastic slope ν_t:

$$y_t = \mu_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2)$$

$$\mu_{t+1} = \mu_t + \nu_t + \xi_t, \qquad \xi_t \sim \mathcal{N}(0, \sigma_\xi^2)$$

$$\nu_{t+1} = \nu_t + \zeta_t, \qquad \zeta_t \sim \mathcal{N}(0, \sigma_\zeta^2)$$

The state vector is $\boldsymbol{\alpha}_t = (\mu_t,\ \nu_t)^\top$ with:

$$\mathbf{T} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{Z} = \begin{pmatrix} 1 & 0 \end{pmatrix}, \quad \mathbf{R} = \mathbf{I}_2, \quad \mathbf{Q} = \begin{pmatrix} \sigma_\xi^2 & 0 \\ 0 & \sigma_\zeta^2 \end{pmatrix}$$

Special cases: setting $\sigma_\xi^2 = 0$ yields the integrated random walk (Holt's linear exponential smoothing); setting $\sigma_\zeta^2 = 0$ yields the random walk with drift.

Example 3: Structural Time Series with Seasonality (Airline Passengers)

The basic structural model (BSM; Harvey, 1989) decomposes a seasonal series into:

$$y_t = \mu_t + \gamma_t + \varepsilon_t$$

where μ_t is the local linear trend, and γ_t is a stochastic trigonometric seasonal component built from harmonic pairs. For a period-s seasonal, we include $\lfloor s/2 \rfloor$ harmonics:

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$$\gamma_t = \sum_{j=1}^{\lfloor s/2 \rfloor} \gamma_{j,t}$$

Each harmonic evolves as:

$$\begin{pmatrix} \gamma_{j,t+1} \\ \gamma^*_{j,t+1} \end{pmatrix} = \begin{pmatrix} \cos\lambda_j & \sin\lambda_j \\ -\sin\lambda_j & \cos\lambda_j \end{pmatrix} \begin{pmatrix} \gamma_{j,t} \\ \gamma^*_{j,t} \end{pmatrix} + \begin{pmatrix} \omega_{j,t} \\ \omega^*_{j,t} \end{pmatrix}$$

where $\lambda_j = 2\pi j/s$ is the frequency of the $j-$th harmonic and $\omega_{j,t}$, $\omega^*_{j,t} \sim \mathcal{N}(0, \sigma^2_\omega)$.

This trigonometric specification has the advantage that it requires only $\lfloor s/2 \rfloor + 1$ parameters regardless of s, versus $s - 1$ parameters for dummy variable seasonality.

The model is estimated on the log-transformed `AirPassengers` series to handle the multiplicative (proportional) seasonal pattern.

<u>Example 4: Dynamic Linear Regression (Time-Varying Coefficients)</u>

A key application of the SSM framework is time-varying parameter (TVP) regression, where regression coefficients are allowed to evolve over time as random walks:

$$y_t = \mathbf{x}_t^\top \boldsymbol{\beta}_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \sigma^2_\varepsilon)$$

$$\boldsymbol{\beta}_{t+1} = \boldsymbol{\beta}_t + \boldsymbol{\eta}_t, \qquad \boldsymbol{\eta}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q})$$

Here $\mathbf{x}_t \in \mathbb{R}^k$ is a vector of predictors and the state vector is $\boldsymbol{\alpha}_t = \boldsymbol{\beta}_t$. The observation matrix $\mathbf{Z}_t = \mathbf{x}_t^\top$ is now time-varying and equals the predictor vector. The diagonal elements of $\mathbf{Q}$ control how rapidly each coefficient can drift; setting $q_{ii} = 0$ fixes the corresponding coefficient as a constant.

This model captures structural change without specifying breakpoints a priori and nests ordinary least squares as a special case ($\mathbf{Q} = \mathbf{0}$).

<u>Example 5: Multivariate State Space Model (DLM Package)</u>

The `*dlm*` package implements the Dynamic Linear Model (DLM), which is equivalent to the SSM. Here we fit a multivariate local level model to two correlated economic series. Let $\mathbf{y}_t \in \mathbb{R}^2$:

$$\mathbf{y}_t = \boldsymbol{\mu}_t + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{H})$$

$$\boldsymbol{\mu}_{t+1} = \boldsymbol{\mu}_t + \boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q})$$

The key difference from the univariate case is that $\mathbf{H}$ and $\mathbf{Q}$ are now 2×2 covariance matrices, allowing for contemporaneous correlation both in the observation noise and in the state shocks.

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Model Selection and Diagnostics

The log-likelihood of the SSM can be computed recursively from the Kalman filter innovations. Given the Gaussian assumption, the conditional distribution of $\mathbf{y}_t$ is:

$$\mathbf{y}_t \mid \mathbf{y}_{1:t-1} \sim \mathcal{N}(\mathbf{Z}_t \mathbf{a}_{t|t-1},\ \mathbf{F}_t)$$

The prediction error decomposition log-likelihood is:

$$\ell(\boldsymbol{\theta}) = -\frac{1}{2} \sum_{t=1}^{T} \left[\log |\mathbf{F}_t| + \mathbf{v}_t^\top \mathbf{F}_t^{-1} \mathbf{v}_t \right] - \frac{pT}{2} \log(2\pi)$$

where $\boldsymbol{\theta}$ denotes all unknown hyperparameters (entries of $\mathbf{H}$, $\mathbf{Q}$, *etc.*). This can be maximized numerically using standard optimizers (*e.g.*, `*optim*` in R). The diffuse likelihood variant handles unknown initial states correctly.

For model comparison, use:

$$\mathrm{AIC} = -2\ell(\hat{\boldsymbol{\theta}}) + 2k$$

$$\mathrm{BIC} = -2\ell(\hat{\boldsymbol{\theta}}) + k \log T$$

where k is the number of estimated hyperparameters. The BIC penalizes complexity more heavily and is generally preferred for large T.

Under correct specification, the standardized innovations $e_t = v_t / \sqrt{F_t}$ should be an *i.i.d.* standard normal sequence. The following diagnostics test this:

Test	Null hypothesis	R function
Ljung–Box	No serial correlation in e_t	*Box.test(e, lag = h, type = "Ljung-Box")*
Jarque–Bera	$e_t \sim \mathcal{N}(0,1)$	*tseries::jarque.bera.test(e)*
Goldfeld–Quandt	Homoscedasticity of e_t	*lmtest::gqtest(lm(e ~ 1))*
Cusum	Parameter stability	*strucchange::efp(e ~ 1, type = "Rec-CUSUM")*

Several numerical issues arise in practice:

1. Loss of positive definiteness: The covariance matrix $\mathbf{P}_{t|t}$ can lose symmetry or positive definiteness due to floating-point errors. Use the Joseph form or square-root filtering (Bierman, 1977), which propagates the *Cholesky* factor of $\mathbf{P}$ instead of $\mathbf{P}$ itself.

2. Near-singular $\mathbf{F}_t$: The innovation covariance matrix may be nearly singular when observation noise is very small. Add a small jitter $\epsilon\mathbf{I}$ to $\mathbf{F}_t$ before inversion, or use a pseudo-inverse.

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3. Diffuse initialisation: When the initial state is unknown, use $\mathbf{P}_1 = \kappa \mathbf{I}$ with $\kappa \to \infty$, handled in practice via the exact diffuse initialization of Koopman & Durbin (2003), which is implemented in `KFAS`.

Concluding Remarks

The following summarizes the Kalman filter–smoother–forecasting algorithm:

Key Strengths

- Recursive efficiency: $O(Tm^3)$ computational complexity, far more efficient than batch methods.
- Optimal under assumptions: Minimizes MSE among all linear estimators (*Gauss–Markov*); minimizes MSE among all estimators under Gaussianity.
- Unified framework: ARIMA, exponential smoothing, structural decomposition, and regression with time-varying coefficients are all special cases.
- Natural handling of missingness: Simply skip the update step when y_t is missing.

Limitations

- Gaussian assumption: The filter is sub-optimal for heavy-tailed or discrete-valued observations. Generalized SSMs and particle filters address this.
- Linear assumption: Non-linear dynamics require the EKF, UKF, or particle filters.
- Hyperparameter sensitivity: ML estimates of $\mathbf{H}$ and $\mathbf{Q}$ can be unreliable with short series; *Bayesian* priors (via MCMC or variational inference) help regularise.
- Non-stationarity: The h-step forecast variance grows without bound for integrated processes; this should be communicated clearly in applications.

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